

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 5

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hello and welcome everyone it's September 30th 2023 and we are in physics as information processing discussion 5 should be a fun discussion on a great fifth lecture so Andre if you'd like to give some opening remarks and then also anyone else whether it's your first time joining a discussion or not it would be great to hear your remarks on lecture five or just where you're coming at it today so first though Entre sure hello Daniel um so I may as well start off by sharing the slides right what do you think sure uh so just just uh before that's a warm-up um let's let's review some of the things that Chris mentioned last time um near the beginning the main thing was this distinction between reductionist and scale free theories obviously one one very Salient feature of of um the free energy literature is that uh it falls under scale free theories these are theories where there is no preferred scale of energy or distance reduction its uh theories are based on on such a scale existence right you you want to break the pieces into the fundamental building blocks of the Legos you know people think of usually high energy or particle physics as they you as the archetype of this thing whereas in is in scale three theories uh there is no such thing and and we should we should have the same skeletons or the same framework um at different scales um this is going to be very important I I Chris correct me if I'm wrong but especially for the coming lecture where we talk about biology Chris himself made a few remarks on how the renormalization group flow aka the process of going from from smaller to higher scale maybe trivial in biology that is to say that um there is a self-similarity okay so so some features of biology should remain invariant at different scales that is um what cells are to a body different individuals of a species might be to a to an ecosystem that that sort of idea um So after talking out scale and and perhaps when we get to the question part I guess this is the one thing that I like uh Chris to explain a little more uh more precisely what these diagrams mean I have a rough idea but it would it would be nice to hear him talk about that again uh anyways in the second half basically um we started asking as is the theme in in this in this in this course so far from first principles what do you need to talk about um

space space on time and um there were some theorizing that time might be um more fundamental that is the things that you need for to implement the clock we're thinking about what are the most basic constituents to implement a clock and those may be even more basic than the things you need to implement qrf's or perception of space after all it seems like you need some spatial extension to perceive a space in the first place I can't remember if it was somebody who was tuning in for the lecture or a question afterwards or something like that but I I think this this remark gets to a very deep point somebody I recall mentioned that some organisms it might have increased himself I can't remember really that there are organisms you may know the name Daniel that are so tiny that if they if they try to implement a gradient to try to chase you know their food or whatever they're they're so small that they can't even perceive the gradient you know like in any any attempt at implementing such perception they're gonna feel uh they're gonna feel the same amount of concentration in all directions so they're basically not large enough to perceive space however uh interestingly so they can they can perceive time I don't know how this ties to the remarks that Chris made about space being more fundamental I suppose Chris was getting at the point that you need fewer parts to perceive space and the the fact that such small organisms um compressive time but not space I guess is further empirical evidence for this thesis and um I actually Andrew if I can uh yeah I I proposed earlier that that time is actually more fundamental in space that one has to have time uh to be able to see space so that's further on in these slides uh that issue is discussed yeah so did I do you want to say something about deckers well this is the right slide that yeah one one just needs a memory really to uh to have a clock an ex an external clock I mean to get to have an internal clock when just when one also has to have a memory do I have an internal clock but if you have a memory and there's some kind of uh activity that's uh periodic in the environment then you have a clock an external clock uh but you don't need object identity to and that's how what distinguishes it from space you have to have object of identity uh or in other words motion the ability to perceive motion to perceive space right yeah any periodic behavior is effectively a clock whereas you know for space we need um object persistence which uh two comments that we've made where that object persistence is the or rather Chris made uh is the defining feature of classicality you said that last time is that right Chris and also that object persistence is is precisely the notion of thickness uh the way Christian would put it yeah yeah object persistence is is definitely a classical notion and it it does seem to be the the place where classicality enters quantum theory as far as I can tell is is with this notion of object persistence that that's the most fundamental place that it enters so so how does that

definition of or rather relationship between classicality and quantumness be an object object persistence how does it compare with other accounts coming from from qdar and the coherence and all of that well say Quantum Darwinism is is a good example uh Quantum Darwinism assumes object persistence right it assumes that there is some persistent thing that interacts over time with the environment uh According to some interaction that has fixed eigenvalues so uh Quantum Darwinism already assumes object persistence and it also assumes classical communication as we discussed a couple of times ago in that it's a theory that is about the ability of multiple observers to agree about the state of some some other object which for them to agree about it that object not only does the object need to persist the the various observers need to have the same description of that object or the same name for that object or or the same way of identifying that object so yeah Quantum organism actually assumes a lot of classical features uh the basic theory of Environmental decoherence is in effect a single Observer Theory so it doesn't have to assume classical communication but it does assume object which which obviously you forget to talk about empirical science you have to right you have to assume that your detector is a thing that persists through time or you can't do an experiment and more importantly than that you have to assume that you're a thing that persists Through Time so we need these classical Notions to talk about doing science yeah I'm not the first person to point that out of course I mean Bohr pointed that out in 1928 in this paper in science where he summarizes quantum theory so it's a very very old idea and I guess um right away I have a question that I suspect I may have asked before but maybe we can revisit it with um from the perspective of today's discussion which is and maybe I should pull uh the slide this is actually from session four um so if we're talking about differences between what what the defining feature of classicality is uh to me that bears the question of here right this tqft um diagrammatic notation for the the classical and the quantum channels so so if we're talking about object persistence as the defining feature of classicality and this is the cartoon picture of what goes on on epr experiment uh Chris could you say maybe a few words about how that classical like how how the Symmetry is broken between the quantum and classical channels I suspect it might have been mentioned that these are symmetrical but yet one of the two has to be a Quantum and the other one is a classical channel the quantum Channel mediates some sort of unitary evolution is that correct whereas the classical channel is is like an information redundant redundancy Reservoir which which maybe has to do something with class with object persistence but I I very much speculating at this point so I'm not sure if Chrissy have any comments yeah well an easy way to think about this I think is to

to actually think about a Bell EPR experiment where you have two observers that are separated in space so so you know we you have to assume here the to talk about Bell EPR you have to assume the notion of space to have a laboratory reference frame uh in which the two observers are spatially separated and in that situation then uh the two observers both independently collect data for some amount of time on on multiple replicas of some Quantum state um that's produced by a source so you have pairs of photons or pairs of electrons or pairs of whatever and at some point the data collection stops and at that point The Observers have to compare their results so it's the it's the comparison step that requires a classical Channel and uh if you if you think about how this happens in practice how it happens via uh a file sent over the Internet or a [Music] piece of paper on which the data are written that's handed by one of the experimenters to the other experimenter or it happens by the experimenters talking to each other so so one experimenter encodes a bunch of a bunch of information in in the ambient air and the other experimenter hears it uh or it could be communicated by light signals right but it's always communicated by some medium and uh that's what defines classical communication it's communication via a medium that persists through time in the observer's reference frame the the minimum time of the Persistence of that Medium is given by the speed of light so if The Observers are a foot apart and they're communicating by a light signal then it may only last in nanosecond oh but if they're talking to each other you know it lasts several seconds and if they're if they exchange a piece of paper and then that piece of paper May last for years in a lab notebook someplace there's always the question of persistence over time that is deeply bound up with this notion of classicality as a process that takes time information has to flow has to be encoded in a medium and then read from that Medium in some finite amount of time in a sense we're assuming something like that in the quantum description when we think of this boundary via which the A and B the the physical components interact oh if we think of that boundary as a qubit array then each of those qubits persists and that's what allows them to encode classical information so encode observational outcomes that are repeated in time that's what allows them to serve as a memory to to encode information in a way that's thermodynamically irreversible and if you think of the boundary in that way as the qubit array then it has a certain thickness in time it's it's it's two plank times thick right the plank time is 10^{-43} seconds so it's it's unmeasurably small but the operation of interacting with the qubit uh is a physical interaction that uh relativistic point of view has to take this minimal time so we're it's it's in a sense time persistence for times that are significantly longer than the plank time that is the marker of classicality here but but even these plank time even even

persistence over the plank time is a necessary ingredient for talking about communication even even Quantum communication so it's a very very deep concept with a with a very long philosophical history going back to you know the ship of Theseus and all that people have been wondering about what does it mean to say that something persists Through Time just anyone want to make a comment on that or bring in a new question please feel free to if you haven't brought up anything yet um sure I can I can speak more I mean um if you don't mind uh I also recall uh and Chris correct me if I'm wrong we we said that related to what you just said that uh clock tick is always tantamount to one of this um classical channels is that is that correct did I remember that correctly Chris yeah uh an internal clock corresponds to an irreversible right on the boundary um so one can then ask what does it mean to say that that's irreversible what it means is it took some energy to do it if you go back to to read that bit again there's at least a reasonably high probability that it's still is in the same state that you wrote it in right and clearly that probability is going to degrade through time because that action is on the boundary so whatever you've written whatever bits that you've written are now exposed to the environment so if Alice writes that they're now exposed to Bob and to Bob's Dynamics as As Time Goes On by your internal clock or any external clock the probability of that memory being the same when you go back to read it slowly decreases or maybe it decreases fast and you know we know that from our own case our own memories degrade Through Time uh because the physical world is interacting with our brains in ways that we don't control you know killing neurons are changing their activity or whatever so whenever we talk about memory uh it's always with this caveat that the the memory is exposed to the environment and it's not guaranteed to be the same it's it's only kind of a great grace of the environment being reasonably benign that any of us have memories at all and and so um so this thermodynamic cost to to classical information sharing or or writing um this in a sense would it be fair to say that in a sense the price you have to pay to to protect against the coherence or you know and entangling this sector with a with the rest of the environment um just to preserve its its identity would that be fair to say oh yeah I mean you can think of decoherence as a good thing or you can think of decoherence as a bad thing right if you're if you're trying to preserve a Quantum State then it's a bad thing you have to spend energy to prevent decoherence by you know cooling everything down to millikelvins oh but if you're interested in preserving classical data then decoherence is a good thing because it prevents your classical data from being completely rearranged by becoming entangled with the environment so I mean think about trying to to write classical memories in using the states of

atoms in the middle of the sun and the temperatures so high that uh you can't do anything you can't rearrange the atoms in any way and expect them to stay put there you're in effectively a completely Quantum environment just because of the high temperature so it's it's a plasma yeah and then just one one final question that popped to my mind as you say this and I can certainly leave it here for everyone else so if we're trying to protect about against the random um lawsuits against the loss of information uh I suppose it really backs the question of um what is the source of this noise that can perturb the bits in the binary um with there's there's obviously this this distinction between a thermal and Quantum noise you know again it's a stretch to see if we can think back about the weak rotation about that but um again I recall earlier sessions where we talk how the noise here in this Quantum theoretic picture of information uh processing is coming from the QRS failing to be aligned um then you know you hear about Notions of quantum noise versus thermal noise uh Chris do you have any comments on all of that um yeah I think an important thing here let me back up yeah the notion of noise like so many other Notions very often uh treated as objective as Observer independent and so when we're doing classical physics we think of noise as just you know there's there's some objective Randomness in the universe and we call that noise so in this picture because we're working with just a bipartite decomposition right we just have Alice and Bob and that's it that's the entire universe um there's there's no external environment here to impose any sort of noise but since we only have two systems of Alice and Bob equals everything else you know not Alice then uh if we talk about Bob being a noisy environment that's clearly relative to this boundary we've drawn between Alice and Bob so so noise isn't Observer independent what's what's noise for Alice is just Bob's Dynamics right so if if you think about where these systems come from you've got some universe is in some pure State because it's an isolated system that state is evolving through time which means it's just undergoing phase rotation so there's no noise in that system right it's perfectly unitary etc etc as soon as you draw a boundary then there's information that the system on one side of the boundary doesn't have about the system on the other side the individual systems are their states can now be considered conditionally independent I mean you can you can come up with a description in which their states are conditionally and that just means that what the other side is doing uh looks like noise unless uh Alice say can come up with a way of analyzing the data that she sees on her boundary uh in a way that makes it make sense right and then it's then it's then it's do we say it's information for Alice but what does that mean it just means it's behavior on her boundary that she's smart enough to assign some kind of semantics to and if she's not

smart enough to do that then it's just noise so as soon as you make the decision you make the notion of noise Observer dependent it it turns into something that makes a lot more sense mm-hmm and and just to really finish this uh I guess it's been it's been a couple months at least but I think it was back in July that we we talked about the distinction between what you're saying which is uh things having the appearance of noise just because you you have finite resources so you cannot make sense of Bob's internal Dynamics as opposed to um in one of in that cartoon of the of the four possibilities for qrf alignments right some of them were uh thermal in the sense that one one side was was acting on a on a boundary acute while the other one was none so the then it truly looks like like noise to the other side um does that does that get to the bottom of the distinction between Quantum and thermal noise at all Chris or am I misunderstanding things here yeah it yeah it turns the distinction into essentially a relative qrf relative distinction right right if if uh you know I mean languages provide a good example I mean think of going to some place where you don't understand the language at all like you know Papua New Guinea or I certainly wouldn't understand the language that that language the behavior is speaking that language encodes lots of information for those people and it codes no information at all for me so it looks like noise to me I can't make sense of it but it's not always from their point of view now let's think about this need to have a thermodynamic sector if you're a an organism that has to burn energy or any system that has to burn energy to to to act irreversibly on your environment of that thermodynamic sector looks to you like noise but it's it's just temperature Heat and uh you use that heat to drive your processing and if you're a really good heat engine then you can use it pretty efficiently and get you know a lot of bang for the buck in terms of information processing if you're really inefficient heat engine maybe it you have to use almost your entire environment as heat to do any little bit of information processing at all so that's not a statement about the universe as a whole that's the statement about you and and your information processing capabilities in your in your thermal your heat processing capabilities you know how how good is your Carnot cycle determines how the amount of information processing it can do thank you Chris um Ian hi uh yeah thank you everyone for putting on this series it's the first time I've joined the discussion sessions yeah so it's a question about in one of the slides there was um a green rectangle and we were talking about that being a or Chris was talking about that being a kind of um thinking of that as a bacterium um and it got me wondering whether like how how else I could get an intuitive grasp of this kind of space-time emerging so what I was thinking is you know it appears like to us humans it appears like the universe is expanding um through time and I was just trying to

think you know a bacterium if we placed a single bacterium on a petri dish um at Time Zero and or even something more simple you know let's say deep sea hydrothermal vent where life was supposed to be you know these agents were supposed to have maybe begun existing as um phospholipid type molecules so I was just wondering you know if I was if I was a phospholipid within that bacterial membrane um just a single one single phospholipid and all I can measure is the existence of phospholipidus or not phospholipidus or and selfness and not selfness then you know I was just trying to imagine as this bacteria divides after 20 minutes let's say the average sort of bacterial division time then before that happened all I could tell was I'm surrounded by phospholipinus so there was no real distinction in my environment but once um the bacteria is divided into two then I'm sitting then I'm observing some phospholipinus which isn't in my sort of what I call my own boundary um so then I've kind of created space space and time is starting to appear even though the petri dish hasn't changed in size or my you know the universe hasn't really changed but it's the phospholipids the orientation of the phospholipids so yeah I kind of I was just trying to liken it if this is a scale invariant principle how can we liken it to how other things might experience space and time emerging that's where I was coming from am I kind of totally barking up the wrong tree with that or is there is that no I I think that's I think that's a really good question and I I think you're you're constructing a very good kind of thought experiment and it's it's this sort of question that I'm really going to try to address in October how how can we think about the fep is a scale theory in practice you know as as applying at the level of of macromolecules as you were just describing you know how do we come up with a description of what a phospholipid say can actually measure uh what actually affects its state and uh you know to what extent does it have a self-representation my guess is that a phospholipid probably doesn't but a bacterium seems to have at least some degree of self-representation I mean it has an immune system it can feel stress so it's starting to seem like uh an organism that's not that different from us but even even something like E coli it can certainly behave in what we call space but it's at least not clear to me that it has a representation of space although it it their processes in the E coli that do distinguish say of the two ends of the E coli I mean even if you think of the chemotaxis pathway uh the two ends of that pathway are quite distinct functionally so perhaps E coli has a one-dimensional representation of space it can tell it's anterior end from its posteriorate but perhaps it doesn't have any representation of what we would call left and right or up and down but if you look at multicellular organisms even something like say Hydro how especially quite sophisticated kind of thing I mean it's not bilateral it's cylindrically

symmetrical it's got a very large distinction between anterior and posterior and it seems to know quite a bit at least I understand from people who work on Hydra about um its own body the state of its own body so I think these are exactly the same kinds of questions that we should be asking and they also apply at the scale that's much larger than us so if we think about a human society for example as an agent what can that agent measure and what can that agent represent how complex can that agent's cognition be and I think those questions are all essentially open so yeah it's a great thought experiment I think it addresses exactly the questions that we're interested in addressing here thank you Chris Harris and then anyone else with a raised hands yes uh thank you once again for allowing me to be here with all of you uh interesting as always so I have a question that is related to this notion of of the dual aspect of information basically the meaning of formation is not just a matter of interpretation but rather it is rudely in rooted in the physical properties of the system under investigation so basically that as I understand it means translates that the meaning of information is not separate from the um but it is rather an intrinsic part of it so uh I I am also familiar um with this uh notion of uh from from the theory of computation called then Bremer response limit that is a limit on the maximum rate of computation that can be achieved let's say in a self-cost contained system uh within the material so my question is how can we parse let's say though that that sort of uh computational limit in terms of information processing thank you I am not sure that we can talk about a I think it's very difficult to talk about the the classical information processing capability of the universe as a whole I mean one has to impose some sort of classical description and the question is then where does that description come from and uh from from kind of whose point of view is that a description but certainly as soon as we you partition the universe as a whole into some some system and some other system one can immediately talk about their information processing capabilities then the question Becomes of Quantum information processing capabilities versus classical information processing capabilities if you're if you're looking at classical to the extent that it's not reversible it requires energy and so one can immediately derive limits based on the free energy access of that if you're talking about reversible information processing which which purely Quantum information processing is then you don't have that energetic requirement but you also don't have a well-defined answer right in any classical terms and it's it's only the if if Alice is your system of Interest that's able to do Quantum information processing then it's only her interaction partner Bob who can can use her as a quantum computer and then uh his ability to employ Alice as a quantum computer is limited by his classical information processing capabilities because he

has to define a classical input and a classical output I'll I think this question becomes very complicated as as soon as it is cast in these Observer relative terms yeah but but but such limits clearly exist right they they just require certain assumptions to Define them so Quantum information and indeed we understood us a kind of dual to the classical information right yes it's essentially one of these bulk boundary type dualities that that seem now to be ubiquitous in physics if we think of um I'm just thinking in terms of the holographic principle um you can have Quantum processing going on in in the bulk in in the system that has a boundary and that can encode it can that can read classical information from the boundary and encode classical information on the boundary quantitatively the holographic principle tells you how much right what can the maximum entropy classical entropy of the boundary be what it doesn't tell you is anything about the semantics that either the system inside assigns to the information or any Observer outside assigns to the information it just it's just a quantitative statement so in in any setting that that's characterized by that principle the classical information on the boundary is dual in an interesting sense too the quantum processing that's going on inside uh in that it that classical information on the boundary encodes everything that an external Observer can know about what's going on on the inside thank you yes now okay yeah on it and then Dean after bye thank you very much um I'm also new to um your active inference and your course uh I just want to say that unfortunately I work with humans and uh um but and I say unfortunately because uh the complexities are horrific uh having said that maybe my my question will make sense um so as I'm learning about Quantum information Theory and I'm what I'm trying to do is assess how can this help me answer questions that I have in the systems that I work with my question is in order to have time Arrow irreversible process don't you need to have a goal and so answer my question is does it matter to even worry about goals because as I listen to all the descriptions and I think about Alice and Bob and systems I immediately think about well so much of what we're talking about depends upon the goal of the system and in some very abstract way it seems to me that there is a goal that's driving the need for the boundary the need for the communication um and so um I'm I'm so obsessed with this question that I wanted to ask about ask it early on before I proceed with my learning uh yeah that's that's a very good question I'm you can think of the whole free energy principle as a a way of of describing physical systems in a way that makes their behavior as agents uh obvious and so the the Effie you could State the fep by saying that every system has a gulf and its goal is minimizing prediction error right admit it or to put it positively its goal is to predict and therefore respond to its environments actions in a way that preserves its own identity I don't

know whether you've read much of Carl fritzen's work but he uses the word self-evidencing a lot it says the goal of any system is to uh generate evidence of its own existence which sounds very psychological and Carl interestingly enough studied first as a psychiatrist so I expect that his point of view is actually very constant with uh your question and where you're coming from here um yeah actually I I started oh I'm sorry I didn't need to interrupt no no go ahead uh no I I started I've spent some time reading Chris feisen's um ideas and work on oh the free energy principle and uh self-evidencing and yes I will continue to dig deep uh the problem is so I'm an anthropologist and I've worked with indigenous peoples and spent 20 years of my life living in these small-scale societies and and so one of my observations is that and that that I find really fascinating but we don't have a good understanding of it and that's why I decided to dig into physics as information processing uh which is that goals are constantly being negotiated and it's it's uh in more along the lines uh so Michael Levine's work has actually was what led me to explore Quantum Theory which is you know like this Collective intelligences that are exists within systems and so at each level of collective intelligence it's not just that you have a collective dictating a goal but that those goals are being negotiated uh with as different as the environments change and as relationships between the systems change so I know these These are uh very big questions but if you could just like help me give me some thoughts on what your thoughts have been about these kinds of questions um yeah I think you're pointing to something very important here and uh uh this is uh a question that this kind of free energy principle based work motivates immediately uh when you consider A system that comprises a bunch of of subsystems you know I I work with Mike Michael Levin at Tufts in developmental biology yeah and one of his slogans as all intelligence as collective and that immediately brings up this issue of of how do these components as you put it uh negotiate their goals how do they how do they uh collaborate how do they compete to what extent are they trying to deceive each other or coerce each other or all of us and we're you know trying to apply this in a Cell Biology context as a as a description of development and morphogenesis I mean if you think about cells in a developing organism they're constantly facing new environments uh their capabilities are constantly changing and how they interact with each other is constantly changing and this all seems very psychological or sociological so I I think what you're pointing to is the ubiquitous need for this kind of in areas other than the human Sciences I mean it's language that's developed within the human Sciences it is language that seems to be important at at much smaller scales and you know clearly when I what would anticipate it being important in much larger scales you know the scales of countries are

civilizations or whatever I guess my answer is yes this is an extremely important point of view with very broad implications for how we think about biology at least definitely looking forward to your continued learnings and journey thank you so much Dean yeah my question has changed three times because uh just as many other questions that have come up in the meantime um so some context here Chris you've you've kind of answered this you mentioned dual as in d-u-a-l and then Anna brought up like is there some kind of competition to dual a d-u-e-l going on and how do we put those two things together and me always thumping the table about minimum two we have to have to keep at least two truisms up at once if we're going to gain some sense around things I want to go back to that lock loc C slide if we can for a second because Chris you mentioned your statement was you draw a boundary and to me there's a lot of information packed in those four words you can draw and you can undraw you can bound you can Unbound those are perfectly contextless subject matter information carrying frames but you also have in that packet you and at which are imperfect orientations an indefinite article and all of us occupying a slide here none of us claiming to be anything other than a contradiction so one aspect of that packet two words of it literally half of it is perfectly contextless and is information carrying and the other literally half of that is prediction matter or information graphing it's literally the the byproduct or the the space-time construct that that is included so I guess what I'm wondering is is how do we make these different matters the subject matter and the prediction matter Tango how do we get them to relate because in the session five you went on a detour you said we'll take a detour which is it holds this exact same structure as you draw a boundary so there's perfect information and there's imperfect information in that they're dancing we do this all the time this seems to be at a minimum what information theory is and that's why I want to kind of go back to this slide because this slide was in a much earlier session but it got to session five where the context could be recapitulated and so I'm wondering whether or not we can really talk about Quantum information without the classical or if if there's if that's even possible um well uh I'll go back to Boris Bohr's statement in his night 2028 paper uh he basically said the answer to that question is no uh that uh we're humans we have to communicate with each other we have to do it in language and that's classical communication and if sorry while we are doing that while while we're having this conversation we can't be worried about the question of do I still exist or you know do you still exist we can worry about that question but we can't do it at the same time that we're having this conversation about of the theory of lot protocols or or exchanging data from a bell epr experiment and that's of course deeply paradoxical and I think that's just the situation we're

in he said none of us are claiming to be anything other than a contradiction uh it's it's a good thing that we self-conscious about the fact that we're we're doing something contradictory here or potentially contradictory here a a colleague and I wrote a paper several years ago uh called science generates limit paradoxes and we basically rehearsed a version of this very discussion that we're having right now and the the upshot of that paper is that in any situation in which a science can turn its theoretical apparatus onto its own practice for example when physics can uh turn this analysis of of boundaries onto the practice of physics then it generates paradoxes and those paradoxes have a particular structure which is the structure that a priest I can't remember his first name a logician who works in non uh of dialectic logic calls the limit paradox it you know all of this is coupled to girdles theorems and these other paradoxes of self-reference or or aspects of self-reference but I think this I think this situation really is ubiquitous and inescapable as soon as we applied the tools of science to the practice of science we generate these paradoxical situations in which we have to acknowledge that we are entities of a certain sort we are constructing our theories using a faculty of imagination that lets us act as if we don't have a point of view uh so it lets us act as if what we're talking about is point of view but our our theories themselves tell us that we have points of view and that they're important so we're stuck in this paradoxical situation I I don't know whether that's what you're trying to point to here but it's what um I immediately leap to hearing what you said yeah a lot there anyone else want to add a peace now I'm going to have to go away so thank you Chris we'll stay for a few minutes thank you though this was awesome we're really looking forward to lecture six okay thank you all and thank you all for attending this discussion session it's it's nice to have this kind of real-time discussion which is missing during the actual lecture sessions so I appreciate it thanks again see you thank you thank you Chris ciao Kate do you want to ask anything or anyone can just bring up anything they're thinking right now nothing in really looking for looking forward to the last lecture on the biology of it all great well any other thoughts or questions or if anyone in the live chat has a thought or a question they can write it and we'll read it in the coming minutes but I just wanted to thank you for creating the space um it's been absolutely amazing that I even that I was able to find this situation where people are talking about enormous questions a very difficult questions and and I I just want I just thought I'd share that I um find in Academia there's I don't know why we're not having these conversations and I've been um it's been like a 10-year Journey and I have finally found this group I just wanted to express my gratitude I'll read a question from the live chat and anyone can give their thoughts so Mills 8102 wrote it seems that consilience would

aim to ameliorate the problems of self-reference is this a category of solution that seems valuable or fraught could consilience ameliorate the problems of self-reference yes Dean can I just ask a different question built on that which is what if the contradiction is a feature as opposed to a fraughtness do I contradict myself very well then I contain multitudes there's the Whitman angle then there's the William Blake life is a fiction and is made up of contradiction so what is the problem of self-reference whose problem is it Harris yeah I will try to answer a bit on both questions and by offering my my views from the cognitive science perspective basically I think it has to do with consilience has to do with the numerous cognitive bias that we all we all are let's say as uh subjected to so there is a very long list of cognitive biases and there are more than 100 listed I think this is one of the challenges in terms of achieving let's say a good levels of conciliency across various domains so this is what I wanted to add another thought on that is consilience though it can mean many things uh synthesis oh yeah please honor go ahead oh yeah I just so I don't really have insights into the self-referencing problem I need to do a lot of thinking about that um but coming from training in evolutionary biology spending you know graduate school post-graduate school working on explanations where we thought of many hypotheses models Etc based on a goal which is organisms are designed to maximize their Fitness and of course her lots of we know that there are lots of problems with that and it turns out with our data what we found is that was not the case uh in our in the populations that we're working with too often the goal did not seem to be Fitness so but what God experienced KB is that there was something we could hang on to in order to uh allow ourselves to propose models for why humans behave the way they do and the decisions that they make um so that's why I go back to goals if if at whatever level of collective intelligence we're talking about there is a goal but even in Harris please let me know if I'm this will be correct that even cognitive biases exists they exist because there is some goal and therefore we can talk about cognitive biases with respect to some goal yes there is some sort of teleonomy in conscious agents or cultural systems as I call them and it has to do with with course yes cognitive biases and and the limitations on the knowledge let's say maybe it's limitations on parts of the knowledge representation on that each of us has and maybe there are limitations in terms of access to to knowing and there are various forms of knowing so it depends it depends on many things and for example one of the cognitive buyers that I believe can have a negative impact on on conciliency let's say in in addressing the this self-referentiality will be the halo effect so where you seem to prioritize let's say or give some advantage to some types of individuals ideas or even theories because of that hello or

whatever reason that is paid it is based on maybe on many different things maybe on your past history or other things so thank you Anna and um if I may add something uh rather briefly um your question reminded me of um Levins um and I think it was written with Phil uh their proposal about the reduction of stress as a goal that con that guy is the emergence of collective structures is one of the papers uh formulated that um that that seems to be the goal that drives things as they grow in complexity reduction of collective stress or reduction of stress at any level yeah I don't remember the title of the paper but that was them just to identify understood it correctly thank you I'll look that up great another piece that of the Levinson Fields collaboration that Chris alluded to earlier is that the notion of computation is Observer dependent that's poly computation if there was a fully encrypted computation an observer without the key would only be seeing noise to them whereas if they were to have the decryption key then they could fully understand and go into more detail about the computation so let's say that we're in relation or observing or investigating some life phenomena at some scale but we're speaking the scale free setting but we're observing a thing that's there people have ascribed different goals to describe what we call goal-oriented Behavior like seemingly to survive or assumingly to have evolutionary Fitness however one an observer could also project different proximate goals like the goal to minimize stress the goal to minimize free energy the goal to minimize surprise those proximate Observer proposed goals as if goals may be exactly isomorphic or aligned with Fitness maximization so that could just be multiple perspectives of projective goals and whatever the Observer projects that doesn't mean it describes the the actual process of the phenomena at all there's just nothing Alice can make any story about Bob and it it may or may not have anything to do or even be possible to be of the kind that Bob tells about himself necessary I want to just piggyback on that Daniel this is really huge and important the moment you use the word goals or the moment you use the word Fitness you are assuming context whereas a lot of this stuff that's happening on this holographic screen is completely abstractless abstracted and completely context less there's no no context required so if you're going to focus on that part of it as long as you can juggle both at the same time and keep the two things apart because one does not require context the other completely assumes that context is given and now you're taking that into your Calculus right but they're not the same thing so I'm just I want to reinforce what Daniel just said thank you Dean Ian then continue thank you yeah I was just maybe reinforcing what you were saying or reflecting on it I was thinking about the behaviors we see in other animals and even ourselves that we might not think are related to Fitness like I don't know there's those kind of um gazelle

type creatures that just jump into the air and like but they seem to do that at a time which might make them um predated upon by a some tiger or lion you know it's you have to then all the I'll just try and figure out how is this related to Fitness um we may do the same you know if I go and mow my lawn waste half a Sunday afternoon mowing my lawn what's that got to do with my fitness well actually it's to do with keeping up with the Joneses and um my social status so um it's really hard to actually know what my behavior is about I can create a story to say that it's to do with uh minimizing free energy or Fitness but really is it or not I don't know is that kind of what you were getting out or something different yeah that's great playing also it's it's come up several times about goals the term goal is not to be found within the active inference ontology strictly speaking that's why we have so much fun with what we expect and what we prefer how can it be what we expect and be the positive thing that we pursue well that's how Paths of least action work and then if somebody else wants to look at that path of least action and say well here was a mile point and this is where it looked like they were heading north and this is where it looked like the goal was to evade this those are all stories that an observer could tell about a path or that the path might tell itself about its own path but those goal narratives don't need to be implemented proximally or primally with goals as a kind of core player in the active inference ontology I'll read some of these great comments and questions just if anybody has a thought on them from the live chat just feel free John Reed wrote what does the active inference group I think that's us but includes everyone think about the equation of pilot wave with many worlds recently expounded upon by Chris O'Dowd at PBS I found it to be mixing terms and frustrating personally any thoughts on pilot wave and many worlds and or I would also just be curious on like the physics or technical side where does what Chris is sharing connect to these kinds of discussions that people have about many worlds and different interpretations of quantum uh yeah that's a great question so I I don't know that I can answer it I know that uh when it comes to interpretations of quantum mechanics whether you know the collapse or the many worlds or whatever I I think what they're hinting at is an increase would probably answer something along those lines that quantum theory is fundamentally about information and until until we make sense of that um fully and and and fully spell out all the details I think this philosophical issues are the interpretations of quantum mechanics are going to linger I am I don't have any strong thoughts uh on interpretations and I would I would say I'm a an agnostic about about interpretations of quantum mechanics I I don't I don't know that I'll subscribe to any of them I don't have the slightest uh idea about cool hairs regarding pilot wave I was conversing with chat GPT and I

actually asked what did uh thought about a pilot wave after watching a lecture on it and then I got in tricked and I wanted to ask and this actually happened the the model started replying and I got the chance to read a bit the the reply and the reply was like it is very interesting but most probably is not wrong is not correct it is wrong and then it must have reached some sort of information goal it retracted the the reply and is replied to me can you ask me a different question so this is what uh this is my short experience with pilot wave theory yeah ask a question is a pilot wave more or less than 100 feet in height so yes exactly exactly 100 any other thoughts or questions it's a fun and Lively live chat too which is always fun yeah well we are um coming right along in the year and now 5 6 completed this course any comments that anyone here just wants to make in a closing route anybody who who wants to can go for it um yeah I have a question if if questions come up have come up during the session yes when could I who could I send them to and how great question in the video description you can get to the syllabus of the course and one of the pages on the course website course syllabus is a question and answer form so you'll see several dozen questions that people have already written and Chris gives answers and then it's publicly posted all of the lectures discussions and questions and answers will be published after the course so every question no matter from where or where to all of these questions are encouraged it only makes it a richer and more relevance publication okay Ian Harris or Dean yes Susan also says post and Discord great point yeah I just wanted to add a small Point as a conclusion from my personal personal perspective I really find that the ideas or the ideas of Chris fields they can be revolutionized a revolutionary let's say in in a variety of and different fields fields and I can see it in in Quantum cryptography and also about the what I've mentioned before in terms of quantum knowledge representation so these are the two fields that I can clearly exceed so it is great to be able participate in this conversation so once again thanks to everyone great to participate in this conversation or Quantum knowledge representation whichever you prefer enter any last comments uh not really just looking forward to the next one I guess next year this is going to be an ongoing thing we'll come back stronger next year for okay what a ride thank you all see you next bye bye everyone